

ASTORIA TONGUE POINT, OR, USA

WMO: 994011

Lat: 46.2074N

Lon: 123.7684W

Elev: 7

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 30.2	(c) 32.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 28.8	(k) 51.3	(l) 26.1	(m) 52.6	(n) 5.1	(o) 350	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.9	(c) 74.3	(d) N/A	(e) 70.5	(f) N/A	(g) 68.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.8	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.4	(b) 17.1	(c) 14.9	(e) 27.2	(f) 86.0	(g) 4.4	(h) 3.9	(i) 24.1	(j) 88.8	(k) 21.6	(l) 91.1	(m) 19.1	(n) 93.3	(o) 15.9	(p) 96.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	43.7	43.3	45.7	48.4	53.1	57.4	61.3	62.4	60.9	55.0	48.1	43.0
	DBStd	8.21	4.16	4.66	4.17	3.67	3.83	3.35	3.13	2.90	3.93	3.97	4.96	5.34
	HDD50	934	198	190	145	71	9	0	0	0	0	6	90	225
	HDD65	4814	661	609	599	499	370	230	122	90	134	310	507	683
	CDD50	1632	2	2	12	22	106	223	350	386	328	162	33	6
	CDD65	35	0	0	0	0	1	3	8	11	12	0	0	0
	CDH74	137	0	0	0	2	16	11	29	35	43	1	0	0
	CDH80	25	0	0	0	0	3	1	6	9	6	0	0	0
Wind	WSAvg	5.8	6.1	5.4	5.7	5.9	5.8	6.0	6.1	5.8	5.3	5.3	6.1	6.0
Precipitation	PrecAvg	82.2	12.6	8.5	9.3	6.4	3.9	2.9	1.0	1.4	2.8	7.9	12.7	12.9
	PrecMax	103.2	26.3	21.4	18.8	12.3	6.7	5.7	3.6	4.6	10.9	17.1	26.6	23.0
	PrecMin	59.3	5.2	1.1	1.3	2.5	0.6	0.5	0.1	0.0	0.0	1.4	3.8	6.1
	PrecStd	13.7	4.3	4.4	3.8	2.6	1.7	1.3	0.8	1.3	2.5	3.9	5.4	4.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.2	63.6	66.4	75.9	74.4	78.9	78.1	80.0	69.6	60.3	59.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	53.4	53.3	56.3	58.6	65.7	68.2	72.1	71.9	72.9	65.1	58.6	55.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	51.6	51.0	53.8	55.5	60.5	64.4	68.6	69.1	68.8	62.2	57.0	52.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	49.8	49.1	51.5	53.2	58.0	62.2	66.4	67.0	66.1	60.2	55.1	49.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.1	9.3	8.7	8.3	7.8	8.6	8.9	9.7	9.2	7.9	7.0
		MCDBR	10.4	10.7	12.4	12.9	14.1	12.4	13.6	13.2	15.4	12.2	8.6	9.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.331	0.318	0.335	0.366	0.388	0.368	0.352	0.358	0.346	0.336	0.326	0.321
	taud		2.478	2.553	2.518	2.392	2.350	2.430	2.503	2.483	2.526	2.548	2.524	2.497
	Ebn at Noon		243	275	285	283	279	285	288	282	277	264	244	234
	Edh at Noon		22	24	29	35	38	36	32	32	28	24	21	19
All-Sky Solar Radiation	RadAvg		353	622	925	1344	1674	1832	1990	1709	1310	769	416	294
	RadStd		54	82	112	120	149	162	154	132	113	88	53	41

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+38

Nomenclature: See separate page